Topic 03: Method of Partial Fractions
Unit V — Inverse Laplace Transforms

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Contents

1 Opening Hook

Hook

After solving an ODE in the s-domain, you often get something like $Y(s) = (3s + 7)/[(s+1)(s^2+4)]$. You cannot invert this directly — it does not match any standard table entry. But if you split it into $A/(s+1) + (Bs+C)/(s^2+4)$, suddenly you have two pieces, each directly invertible: Ae^{-t} and $B\cos(2t) + (C/2)\sin(2t)$. The method of partial fractions is the "divide and conquer" strategy for inverting any rational transform. Master it and you can invert virtually anything.

2 Why This Topic Exists

"Before we begin, here is the honest answer to why you are reading this..."

Every ODE that is solved via the Laplace transform produces an algebraic expression $Y(s) = P(s)/Q(s)$ in the s-domain. In virtually every engineering problem, the denominator $Q(s)$ has degree ≥ 2, so $Y(s)$ does not directly match any entry in the standard Laplace table. Partial fraction decomposition breaks this one complicated rational function into a sum of simple pieces, each of which is a recognisable table entry. Without this technique, inverting the transform is impossible for most practical problems.

Mistake	Consequence
Trying to invert without decomposing first	No table entry matches — stuck completely
Using wrong form for repeated roots	Incomplete decomposition — missing $A_2/(s-a)^2$ terms
Treating complex roots incorrectly	Getting complex-valued $f(t)$ instead of real sinusoidal functions

Key Takeaway

Partial fractions are the universal bridge between a complicated $F(s)$ and the standard Laplace table. There is no shortcut — you must decompose first.

3 You Already Know This (Intuition First)

Think of breaking a complex fraction in ordinary arithmetic:

$$\frac{7}{12} = \frac{3}{4} - \frac{1}{6}.$$

You replaced one "hard" fraction with two "easy" ones that you already know how to handle. Partial fractions do exactly the same for rational functions of s:

$$\frac{P(s)}{Q(s)} \longrightarrow \frac{A_1}{s-r_1} + \frac{A_2}{s-r_2} + \cdots + \frac{A_m}{(s-r)^m} + \cdots + \frac{Cs+D}{s^2+as+b}.$$

Each piece on the right maps directly to an entry in the inverse Laplace table: $A/(s-r) \to Ae^{rt}$, $A/(s-r)^2 \to Ate^{rt}$, $(Cs+D)/(s^2+as+b) \to$ a shifted sinusoid. The entire strategy is: break it apart, then invert the pieces.

4 Prerequisite: Proper Rational Functions

Proper vs. Improper Rational Functions

Let $F(s) = P(s)/Q(s)$.

- $F(s)$ is **proper** if $\deg P < \deg Q$.
- $F(s)$ is **improper** if $\deg P \geq \deg Q$.

If $F(s)$ is improper, perform **polynomial long division** first to write $F(s) = G(s) + R(s)/Q(s)$, where $G(s)$ is a polynomial and $R(s)/Q(s)$ is now proper. Then apply partial fractions to $R(s)/Q(s)$ only.

Example: $F(s) = \dfrac{s^3 + 2s}{s^2 + 1}$.
Divide $s^3 + 2s$ by $s^2 + 1$:

$$s^3 + 2s = s \cdot (s^2 + 1) + s \implies F(s) = s + \frac{s}{s^2 + 1}.$$

The polynomial part s inverts to $\delta'(t)$ (derivative of the Dirac delta). The proper part $s/(s^2 + 1)$ inverts to $\cos t$ directly from the table.

Degree condition	**Action**	**Example**
$\deg P < \deg Q$ (proper)	Apply partial fractions directly	$\dfrac{3s+7}{(s+1)(s-2)}$
$\deg P = \deg Q$ (improper)	Divide once, then partial fractions	$\dfrac{s^2}{s^2+1}$
$\deg P > \deg Q$ (improper)	Divide until proper remainder	$\dfrac{s^3+2s}{s^2+1}$

5 Case 1 — Distinct Real Roots

Decomposition Form: Distinct Real Roots

If $Q(s) = (s - r_1)(s - r_2)\cdots(s - r_n)$ with all r_i distinct reals:

$$\frac{P(s)}{Q(s)} = \frac{A_1}{s - r_1} + \frac{A_2}{s - r_2} + \cdots + \frac{A_n}{s - r_n}.$$

Heaviside Cover-Up Method:

$$A_k = \lim_{s \to r_k} (s - r_k)\, F(s) = \frac{P(r_k)}{\prod_{j \neq k}(r_k - r_j)}.$$

Multiply both sides by $(s - r_k)$, then set $s = r_k$ to "cover up" that factor and evaluate directly.

5.1 Worked Example — Case 1

Example: $F(s) = \frac{3s+7}{(s+1)(s-2)(s+3)}$

Write the decomposition:

$$\frac{3s+7}{(s+1)(s-2)(s+3)} = \frac{A}{s+1} + \frac{B}{s-2} + \frac{C}{s+3}.$$

Step 1 — Find A (cover up $(s+1)$, set $s=-1$):

$$A = \frac{3(-1)+7}{(-1-2)(-1+3)} = \frac{4}{(-3)(2)} = -\frac{2}{3}.$$

Step 2 — Find B (cover up $(s-2)$, set $s=2$):

$$B = \frac{3(2)+7}{(2+1)(2+3)} = \frac{13}{(3)(5)} = \frac{13}{15}.$$

Step 3 — Find C (cover up $(s+3)$, set $s=-3$):

$$C = \frac{3(-3)+7}{(-3+1)(-3-2)} = \frac{-2}{(-2)(-5)} = -\frac{1}{5}.$$

Step 4 — Verify at $s=0$:

$$\text{LHS at } s=0: \frac{3(0)+7}{(0+1)(0-2)(0+3)} = \frac{7}{(1)(-2)(3)} = -\frac{7}{6}.$$

$$\text{RHS at } s=0: \frac{A}{0+1}+\frac{B}{0-2}+\frac{C}{0+3} = \frac{-2/3}{1}+\frac{13/15}{-2}+\frac{-1/5}{3} = -\frac{2}{3}-\frac{13}{30}-\frac{1}{15} = -\frac{35}{30} = -\frac{7}{6}.\checkmark$$

Numerical check confirms $A=-2/3$, $B=13/15$, $C=-1/5$.

Step 5 — Invert using $\mathcal{L}^{-1}\{1/(s-a)\} = e^{at}$:

$$f(t) = -\frac{2}{3}e^{-t} + \frac{13}{15}e^{2t} - \frac{1}{5}e^{-3t}, \quad t \geq 0.$$

What Did We Learn?

The Heaviside cover-up method lets you find each coefficient A_k with a single substitution — no system of equations needed when all roots are distinct. Each resulting term inverts immediately to a real exponential.

6 Case 2 — Repeated Real Roots

Decomposition Form: Repeated Real Roots

If $Q(s)$ contains $(s-r)^m$ as a factor:

$$\frac{P(s)}{(s-r)^m \cdot R(s)} = \frac{A_1}{s-r} + \frac{A_2}{(s-r)^2} + \cdots + \frac{A_m}{(s-r)^m} + (\text{terms from } R(s)).$$

Finding the highest-power coefficient:

$$A_m = \lim_{s \to r} (s-r)^m F(s).$$

Finding lower-power coefficients: differentiate $(s-r)^m F(s)$ with respect to s and evaluate at $s = r$, or substitute convenient values of s to generate equations for the unknowns.

Inversion table entries:

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-r)^k}\right\} = \frac{t^{k-1}}{(k-1)!}\,e^{rt}.$$

6.1 Worked Example 1 — Case 2

Example: $F(s) = \frac{s+3}{(s-1)^2(s+2)}$

Write the decomposition:

$$\frac{s+3}{(s-1)^2(s+2)} = \frac{A}{s-1} + \frac{B}{(s-1)^2} + \frac{C}{s+2}.$$

Step 1 — Find C (cover up $(s+2)$, set $s = -2$):

$$C = \frac{-2+3}{(-2-1)^2} = \frac{1}{9}.$$

Step 2 — Find B (multiply both sides by $(s-1)^2$, set $s = 1$):

$$B = \frac{1+3}{1+2} = \frac{4}{3}.$$

Step 3 — Find A (multiply through, compare s^2 coefficients): Expand the right side over the common denominator:

$$s + 3 = A(s-1)(s+2) + B(s+2) + C(s-1)^2.$$

Set $s = 0$:

$$3 = A(-1)(2)+B(2)+C(1) \implies 3 = -2A+\frac{8}{3}+\frac{1}{9} \implies -2A = 3-\frac{24}{9}-\frac{1}{9} = 3-\frac{25}{9} = \frac{2}{9} \implies A = -\frac{1}{9}$$

Step 4 — Verify at $s = -1$:

$$\text{LHS} : \frac{-1+3}{(-1-1)^2(-1+2)} = \frac{2}{4} = \frac{1}{2}.$$

$$\text{RHS}: \frac{-1/9}{-1-1} + \frac{4/3}{(-1-1)^2} + \frac{1/9}{-1+2} = \frac{1}{18} + \frac{1}{3} + \frac{1}{9} = \frac{1}{18} + \frac{6}{18} + \frac{2}{18} = \frac{9}{18} = \frac{1}{2}.\checkmark$$

Step 5 — Invert:

$$f(t) = -\frac{1}{9}\,e^{t} + \frac{4}{3}\,t\,e^{t} + \frac{1}{9}\,e^{-2t}, \quad t \geq 0.$$

What Did We Learn?

For a repeated root $(s-r)^m$, you need m separate terms in the decomposition. The highest-power coefficient A_m comes from the cover-up; lower coefficients follow from substitution or coefficient comparison.

6.2 Worked Example 2 — Case 2 (root at $s = 0$)

Example: $F(s) = \frac{2}{s^2(s+1)}$

Write the decomposition:

$$\frac{2}{s^2(s+1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+1}.$$

Step 1 — Find B (cover up s^2, set $s = 0$):

$$B = \frac{2}{0+1} = 2.$$

Step 2 — Find C (cover up $(s+1)$, set $s = -1$):

$$C = \frac{2}{(-1)^2} = 2.$$

Step 3 — Find A (multiply out and compare coefficients):

$$2 = A\,s(s+1) + B(s+1) + C\,s^2.$$

Coefficient of s^2: $0 = A + C \implies A = -C = -2$.
Verify coefficient of s^1: $0 = A + B = -2 + 2 = 0.\checkmark$
Step 4 — Invert:

$$f(t) = -2 + 2t + 2e^{-t}, \quad t \geq 0.$$

What Did We Learn?

Repeated roots at $s = 0$ produce constant and polynomial terms in $f(t)$. The form $A/s + B/s^2$ is essential — using only B/s^2 would give an incomplete decomposition and a wrong answer.

7 Case 3 — Complex Conjugate Roots (Irreducible Quadratic)

Decomposition Form: Irreducible Quadratic Factor

If $Q(s)$ contains an irreducible quadratic $(s^2 + as + b)$ with $a^2 - 4b < 0$:

$$\frac{P(s)}{(s^2 + as + b) \cdot R(s)} = \frac{As + B}{s^2 + as + b} + (\text{terms from } R(s)).$$

Note: The numerator must be $As + B$ (linear), not a constant, because the factor has degree 2.

Completing the square:

$$s^2 + as + b = (s + p)^2 + q^2, \quad p = \tfrac{a}{2}, \quad q = \sqrt{b - p^2}.$$

Numerator splitting:

$$\frac{As + B}{(s + p)^2 + q^2} = A \cdot \frac{s + p}{(s + p)^2 + q^2} + \frac{B - Ap}{q} \cdot \frac{q}{(s + p)^2 + q^2}.$$

Each term now matches the shifted cosine/sine table entries:

$$\mathcal{L}^{-1}\left\{\frac{s + p}{(s + p)^2 + q^2}\right\} = e^{-pt}\cos(qt), \qquad \mathcal{L}^{-1}\left\{\frac{q}{(s + p)^2 + q^2}\right\} = e^{-pt}\sin(qt).$$

7.1 Worked Example — Case 3

Example: $F(s) = \frac{s^2+2s+3}{(s^2+2s+5)(s-1)}$

Write the decomposition:

$$\frac{s^2 + 2s + 3}{(s^2 + 2s + 5)(s - 1)} = \frac{As + B}{s^2 + 2s + 5} + \frac{C}{s - 1}.$$

Step 1 — Find C (cover up $(s - 1)$, set $s = 1$):

$$C = \frac{1 + 2 + 3}{(1 + 2 + 5)} = \frac{6}{8} = \frac{3}{4}.$$

Step 2 — Find A and B (equate numerator coefficients): Multiply through:

$$s^2 + 2s + 3 = (As + B)(s - 1) + C(s^2 + 2s + 5).$$

Expand the right side:

$$(As + B)(s - 1) + \tfrac{3}{4}(s^2 + 2s + 5) = As^2 - As + Bs - B + \tfrac{3}{4}s^2 + \tfrac{3}{2}s + \tfrac{15}{4}.$$

Collect like powers:

$$= \left(A + \tfrac{3}{4}\right)s^2 + \left(-A + B + \tfrac{3}{2}\right)s + \left(-B + \tfrac{15}{4}\right).$$

Equate coefficients with LHS $= s^2 + 2s + 3$:

$$s^2:\ A + \tfrac{3}{4} = 1 \implies A = \tfrac{1}{4}.$$

$$s^0:\ -B + \tfrac{15}{4} = 3 \implies B = \tfrac{3}{4}.$$

Check s^1: $-A + B + \frac{3}{2} = -\frac{1}{4} + \frac{3}{4} + \frac{3}{2} = \frac{1}{2} + \frac{3}{2} = 2.\checkmark$

Step 3 — Complete the square in $s^2 + 2s + 5$:

$$s^2 + 2s + 5 = (s+1)^2 + 4, \quad p = 1,\ q = 2.$$

Step 4 — Split the numerator $As + B = \frac{1}{4}s + \frac{3}{4}$:

$$\frac{\frac{1}{4}s + \frac{3}{4}}{(s+1)^2 + 4} = \frac{1}{4}\cdot\frac{s+1}{(s+1)^2+4} + \frac{3/4 - 1/4}{2}\cdot\frac{2}{(s+1)^2+4} = \frac{1}{4}\cdot\frac{s+1}{(s+1)^2+4} + \frac{1}{4}\cdot\frac{2}{(s+1)^2+4}.$$

Step 5 — Invert:

$$f(t) = \frac{1}{4}\,e^{-t}\cos(2t) + \frac{1}{4}\,e^{-t}\sin(2t) + \frac{3}{4}\,e^{t}, \quad t \geq 0.$$

What Did We Learn?

For an irreducible quadratic factor, always use a linear numerator $As + B$ in the decomposition. After finding the constants, complete the square and split the numerator to match the shifted sine and cosine table entries.

8 Case 4 — Repeated Complex Roots (Advanced)

Decomposition Form: Repeated Irreducible Quadratic

If $(s^2 + as + b)^2$ appears in $Q(s)$, the decomposition includes:

$$\frac{A_1 s + B_1}{s^2 + as + b} + \frac{A_2 s + B_2}{(s^2 + as + b)^2}.$$

After completing the square $(s^2 + as + b) = (s+p)^2 + q^2$, the second-order term has inverse:

$$\mathcal{L}^{-1}\left\{\frac{s+p}{[(s+p)^2 + q^2]^2}\right\} = \frac{t}{2q}\sin(qt)\,e^{-pt}, \qquad \mathcal{L}^{-1}\left\{\frac{q}{[(s+p)^2 + q^2]^2}\right\} = \frac{1}{2q^2}\big(\sin(qt) - qt\cos(qt)\big)\,e^{-pt}.$$

Equivalently:

$$\mathcal{L}^{-1}\left\{\frac{1}{[(s+p)^2 + q^2]^2}\right\} = \frac{1}{2q^3}\big(\sin(qt) - qt\cos(qt)\big)\,e^{-pt}.$$

Brief example (final form only): For $F(s) = 1/(s^2+4)^2$, completing the square gives $p = 0$, $q = 2$. The decomposition is just $F(s)$ itself (already in the required form with $A_1 = B_1 = A_2 = 0$, $B_2 = 1$). Using the formula with $q = 2$, the factor is

$\frac{1}{2q^3} = \frac{1}{2(2^3)} = \frac{1}{16}$:

$$\mathcal{L}^{-1}\left\{\frac{1}{(s^2+4)^2}\right\} = \frac{1}{16}\big(\sin(2t) - 2t\cos(2t)\big), \quad t \ge 0.$$

For general cases with mixed factors, find A_2, B_2 from the highest-power cover-up on the quadratic block, then solve for A_1, B_1 by comparing coefficients.

9 Systematic Procedure Summary

6-Step Algorithm for Partial Fractions

Step 1. Check proper: Is $\deg P < \deg Q$? If not, perform polynomial long division first.

Step 2. Factor $Q(s)$ completely into linear factors $(s - r_i)^{m_i}$ and irreducible quadratic factors $(s^2 + a_j s + b_j)^{n_j}$.

Step 3. Write the partial fraction form based on the root types: each distinct linear factor gives $A/(s-r)$; each repeated linear factor $(s-r)^m$ gives m terms; each irreducible quadratic gives a linear numerator $(As + B)$.

Step 4. Find all coefficients using: Heaviside cover-up for distinct roots, substitution or differentiation for repeated roots, equating coefficients for quadratic factors.

Step 5. Invert each term using the standard Laplace table (completing the square for quadratic terms as needed).

Step 6. Combine all terms to write the final $f(t)$.

10 Coefficient-Finding Methods Summary

Method	When to Use	How It Works
Heaviside cover-up	Distinct linear roots	Multiply by $(s - r_k)$ and set $s = r_k$; the other terms vanish
Substitution of convenient s values	Any case, especially repeated roots	Choose s values that simplify algebra and generate equations for unknowns
Equating coefficients of like powers of s	Quadratic/mixed cases	Expand numerator, collect powers s^n, match to original numerator coefficients
Differentiation method	Repeated roots	Differentiate $(s-r)^m F(s)$ with respect to s, set $s = r$ to find lower-order constants

11 pgfplots Visualization

For the Case 1 example, with $A = -2/3$, $B = 13/15$, $C = -1/5$, we have $f(t) = -\frac{2}{3}e^{-t} + \frac{13}{15}e^{2t} - \frac{1}{5}e^{-3t}$. Figure 1 shows each component term and their sum on $[0, 2]$.

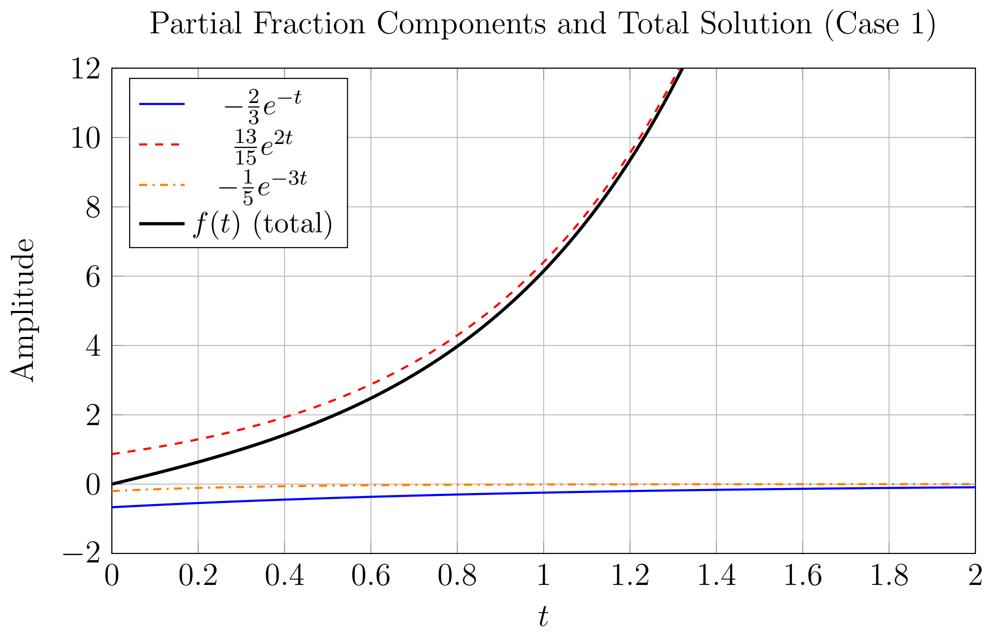

Figure 1: Each partial fraction term contributes one exponential mode — the total solution is a superposition.

12 Worked Examples

12.1 Example 1: Three Distinct Roots

Find $\mathcal{L}^{-1}\left\{\frac{2s^2+5s+3}{(s+1)(s+2)(s+3)}\right\}$

Step 1 — Decompose:

$$\frac{2s^2+5s+3}{(s+1)(s+2)(s+3)} = \frac{A}{s+1} + \frac{B}{s+2} + \frac{C}{s+3}.$$

Step 2 — Find A (set $s=-1$):

$$A = \frac{2(1)+5(-1)+3}{(-1+2)(-1+3)} = \frac{2-5+3}{(1)(2)} = \frac{0}{2} = 0.$$

Step 3 — Find B (set $s=-2$):

$$B = \frac{2(4)+5(-2)+3}{(-2+1)(-2+3)} = \frac{8-10+3}{(-1)(1)} = \frac{1}{-1} = -1.$$

Step 4 — Find C (set $s=-3$):

$$C = \frac{2(9)+5(-3)+3}{(-3+1)(-3+2)} = \frac{18-15+3}{(-2)(-1)} = \frac{6}{2} = 3.$$

Step 5 — Verify at $s=0$:

$$\frac{3}{(1)(2)(3)} = \frac{1}{2}, \qquad \frac{0}{1} + \frac{-1}{2} + \frac{3}{3} = 0 - \frac{1}{2} + 1 = \frac{1}{2}.\checkmark$$

Step 6 — Invert:

$$f(t) = 0\cdot e^{-t} - e^{-2t} + 3e^{-3t} = -e^{-2t} + 3e^{-3t}, \quad t \geq 0.$$

What Did We Learn?

Even when one constant turns out to be zero, the algorithm is the same. The cover-up gives the answer without solving any simultaneous equations.

12.2 Example 2: Repeated Root at Zero and Complex Pair

Find $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s^2+4)}\right\}$

Step 1 — Decompose:

$$\frac{1}{s^2(s^2+4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{Cs+D}{s^2+4}.$$

Step 2 — Multiply through by $s^2(s^2+4)$:

$$1 = As(s^2+4) + B(s^2+4) + (Cs+D)s^2.$$

Step 3 — Set $s=0$: $1 = B\cdot 4 \implies B = 1/4$.
Step 4 — Equate s^0 (already done), s^2, s^3, s^1 coefficients:

$$s^2:\ 0 = B + D \implies D = -\frac{1}{4}.$$

$$s^3:\ 0 = A + C.$$

$$s^1:\ 0 = 4A \implies A = 0 \implies C = 0.$$

Step 5 — Verify at $s = 1$:

$$\frac{1}{1\cdot 5} = \frac{1}{5}, \qquad \frac{0}{1} + \frac{1/4}{1} + \frac{-1/4}{5} = \frac{1}{4} - \frac{1}{20} = \frac{4}{20} = \frac{1}{5}.\checkmark$$

Step 6 — Invert:

$$f(t) = \frac{1}{4}t + \mathcal{L}^{-1}\left\{\frac{-1/4}{s^2+4}\right\} = \frac{t}{4} - \frac{1}{8}\sin(2t), \quad t \geq 0.$$

What Did We Learn?

When the denominator contains both repeated linear and irreducible quadratic factors, use all four terms in the decomposition. Equating coefficients of every power of s gives a straightforward linear system.

12.3 Example 3: Proper Fraction with Mixed Factors

Find $\mathcal{L}^{-1}\left\{\frac{2s^2-s+3}{(s-1)(s^2+4)}\right\}$

Check: $\deg P = 2 < \deg Q = 3$ — proper, no division needed.
Step 1 — Decompose:

$$\frac{2s^2 - s + 3}{(s-1)(s^2+4)} = \frac{A}{s-1} + \frac{Bs + C}{s^2+4}.$$

Step 2 — Find A (set $s = 1$):

$$A = \frac{2 - 1 + 3}{1 + 4} = \frac{4}{5}.$$

Step 3 — Multiply through:

$$2s^2 - s + 3 = A(s^2+4) + (Bs + C)(s - 1).$$

Expand: $= \frac{4}{5}(s^2+4) + (Bs + C)(s-1)$.
Step 4 — Equate s^2: $2 = \frac{4}{5} + B \implies B = \frac{6}{5}$.
Equate s^0: $3 = \frac{16}{5} - C \implies C = \frac{16}{5} - 3 = \frac{1}{5}$.
Check s^1: $-1 = -B + C = -\frac{6}{5} + \frac{1}{5} = -1.\checkmark$
Step 5 — Complete the square ($s^2 + 4 = s^2 + 0\cdot s + 4$, so $p = 0$, $q = 2$):

$$\frac{\frac{6}{5}s + \frac{1}{5}}{s^2+4} = \frac{6}{5}\cdot\frac{s}{s^2+4} + \frac{1}{10}\cdot\frac{2}{s^2+4}.$$

Step 6 — Invert:

$$f(t) = \frac{4}{5}e^t + \frac{6}{5}\cos(2t) + \frac{1}{10}\sin(2t), \quad t \geq 0.$$

What Did We Learn?

For $s^2+4 = s^2+0\cdot s+4$, completing the square is trivial ($p=0$, $q=2$), and the numerator splits cleanly into cosine and sine terms.

13 Excel Example

13.1 Numerical Verification — Case 2 Result

From the Case 2 worked example, $F(s) = (s+3)/[(s-1)^2(s+2)]$, with $A=-1/9$, $B=4/3$, $C=1/9$:

$$f(t) = -\frac{1}{9}e^{t} + \frac{4}{3}t\,e^{t} + \frac{1}{9}e^{-2t}.$$

We numerically verify this by computing the Laplace integral $\int_0^\infty e^{-3t}f(t)\,dt$ using the trapezoid rule and comparing to the exact value $F(3) = (3+3)/[(3-1)^2(3+2)] = 6/20 = 0.3$.

Spreadsheet Setup:

- Column A: t from 0 to 4 in steps of 0.05 (formula in A2: `=A1+0.05`)
- Column B: $f(t)$ exact (formula in B1: `=(-1/9)*EXP(A1)+(4/3)*A1*EXP(A1)+(1/9)*EXP(-2*A1)`)
- Column C: integrand $e^{-3t}f(t)$ (formula in C1: `=EXP(-3*A1)*B1`)
- Column D: cumulative trapezoid sum (D1: `=0`; D2: `=D1+0.05*(C1+C2)/2`; fill down)

Row	A: t	B: $f(t)$	C: $e^{-3t}f(t)$	D: Cumulative Sum
1	0.00	0.0000	0.0000	0.0000
2	0.05	≈ 0.1716	0.1484	0.0065
3	0.10	≈ 0.2546	0.1884	0.0149
4	0.15	≈ 0.3630	0.2306	0.0259
5	0.20	≈ 0.5012	0.2735	0.0396
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
81	4.00	≈ 394.8	≈ 0.0000	≈ 0.2998

What Did We Learn?

The cumulative trapezoid sum converges to $\approx 0.2998 \approx 0.3$, matching the exact value $F(3) = 0.3$. This numerical check confirms that our partial fraction coefficients are correct. Excel provides a quick verification tool for any inverse Laplace result.

14 Python Example

Listing 1: Partial fraction decomposition and inverse Laplace via sympy

```python
import sympy as sp

s, t = sp.symbols('s t', real=True, positive=True)

```

```python
# ---- Example 1: Three distinct roots ----
F1 = (2*s**2 + 5*s + 3) / ((s+1)*(s+2)*(s+3))
F1_pf = sp.apart(F1, s)
print("F1 partial fractions:", F1_pf)
# Expected: -1/(s + 2) + 3/(s + 3)

f1 = sp.inverse_laplace_transform(F1_pf, s, t)
print("f1(t) =", sp.simplify(f1))
# Expected: -exp(-2*t) + 3*exp(-3*t)

F1_check = sp.laplace_transform(f1, t, s, noconds=True)
print("Verify F1:", sp.simplify(F1_check - F1))  # Should print 0
# Expected: 0

# ---- Example 2: Repeated root + complex pair ----
F2 = 1 / (s**2 * (s**2 + 4))
F2_pf = sp.apart(F2, s)
print("\nF2 partial fractions:", F2_pf)
# Expected: 1/(4*s**2) - 1/(4*(s**2 + 4))

f2 = sp.inverse_laplace_transform(F2_pf, s, t)
print("f2(t) =", sp.simplify(f2))
# Expected: t/4 - sin(2*t)/8

F2_check = sp.laplace_transform(f2, t, s, noconds=True)
print("Verify F2:", sp.simplify(F2_check - F2))  # Should print 0
# Expected: 0

# ---- Example 3: Distinct + complex pair ----
F3 = (2*s**2 - s + 3) / ((s-1)*(s**2 + 4))
F3_pf = sp.apart(F3, s)
print("\nF3 partial fractions:", F3_pf)
# Expected: (4/5)/(s-1) + (6*s/5 + 1/5)/(s**2+4)

f3 = sp.inverse_laplace_transform(F3_pf, s, t)
print("f3(t) =", sp.simplify(f3))
# Expected: (4/5)*exp(t) + (6/5)*cos(2*t) + (1/10)*sin(2*t)

F3_check = sp.laplace_transform(f3, t, s, noconds=True)
print("Verify F3:", sp.simplify(F3_check - F3))  # Should print 0
# Expected: 0
```

What Did We Learn?

`sympy.apart` performs symbolic partial fraction decomposition automatically, and `sympy.inverse_laplace_transform` inverts each piece. Taking the forward transform and subtracting the original should give zero — a fully automated correctness check that saves manual verification time.

15 Viva-Style Oral Questions

Q1. When is the method of partial fractions applicable to finding an inverse Laplace transform?

Answer: Partial fractions apply when $F(s) = P(s)/Q(s)$ is a *proper* rational function, i.e., $\deg P < \deg Q$. If $F(s)$ is improper, we first perform polynomial long division to produce a proper fraction, and then apply partial fractions to the remainder.

Q2. State the three main cases in partial fraction decomposition.

Answer: (1) Distinct real roots: each factor $(s - r_i)$ contributes a term $A_i/(s - r_i)$. (2) Repeated real roots: a factor $(s - r)^m$ contributes m terms $A_1/(s - r) + \cdots + A_m/(s-r)^m$. (3) Irreducible quadratic factors: a factor $s^2 + as + b$ (with $a^2 - 4b < 0$) contributes a term $(As + B)/(s^2 + as + b)$.

Q3. Explain the Heaviside cover-up method. When can it be used?

Answer: For distinct linear factors, cover-up says: $A_k = \lim_{s\to r_k}(s - r_k)F(s)$. You multiply $F(s)$ by $(s - r_k)$ (mentally "covering" that factor in the denominator) and set $s = r_k$. It works only for simple (non-repeated) linear factors and gives each constant in a single substitution.

Q4. How do you find the coefficients for a repeated root $(s - r)^m$?

Answer: The highest-power coefficient A_m is found by $A_m = \lim_{s\to r}(s - r)^m F(s)$ (cover-up on the entire repeated block). Lower-power coefficients $A_{m-1}, \ldots, A_1$ are found by either: (a) differentiating $(s-r)^m F(s)$ successively and evaluating at $s = r$ (differentiation method), or (b) substituting convenient values of s and solving the resulting equations.

Q5. What is an irreducible quadratic factor? Why must its numerator in the decomposition be linear ($As + B$) rather than a constant A?

Answer: An irreducible quadratic $s^2 + as + b$ has $a^2 - 4b < 0$, so it has no real roots and cannot be factored over the reals. The numerator must be $As + B$ (one degree below the factor) because a constant A would not provide enough freedom to match the degrees on both sides of the partial fraction identity.

Q6. Describe the completing-the-square step and explain why it is needed.

Answer: We write $s^2 + as + b = (s + p)^2 + q^2$ where $p = a/2$ and $q = \sqrt{b - p^2}$. This is needed because the standard Laplace table has entries $\mathcal{L}\{e^{-pt}\cos(qt)\} = (s+p)/[(s+p)^2 + q^2]$ and $\mathcal{L}\{e^{-pt}\sin(qt)\} = q/[(s+p)^2 + q^2]$. The completed-square form brings the denominator into exactly this pattern.

Q7. How many unknown constants must you find in the decomposition of $P(s)/Q(s)$? How does this relate to $\deg Q$?

Answer: The total number of constants to be determined always equals $\deg Q$. Each simple linear factor contributes 1 constant, each factor $(s - r)^m$ contributes m

constants, and each irreducible quadratic contributes 2 constants. Summing up the contributions from all factors always gives $\deg Q$ in total.

Q8. How does the First Shift (Frequency Shift) theorem interact with partial fractions when inverting a term like $(s+2)/[(s+2)^2+9]$?

Answer: Once we have completed the square and rewritten a quadratic factor as $(s+p)^2+q^2$, the frequency shift theorem directly applies: $\mathcal{L}^{-1}\{G(s+p)\} = e^{-pt}g(t)$. For example, $(s+2)/[(s+2)^2+9]$ is recognised as the Laplace transform of $e^{-2t}\cos(3t)$ without further manipulation.

16 Descriptive Questions

Q1. State the three cases in the method of partial fractions for inverse Laplace transforms and write the decomposition form for each case. Illustrate with an example for each case.

Answer: Case 1 (Distinct real roots): $P(s)/[(s-r_1)(s-r_2)\cdots(s-r_n)] = A_1/(s-r_1)+\cdots+A_n/(s-r_n)$. Example: $(s+1)/[(s+2)(s+3)] = A/(s+2)+B/(s+3)$. $A = (-2+1)/(-2+3) = -1$, $B = (-3+1)/(-3+2) = 2$; inverse: $f(t) = -e^{-2t}+2e^{-3t}$.

Case 2 (Repeated root): $P(s)/(s-r)^m \cdot R(s)$ decomposes with m terms $A_1/(s-r)+\cdots+A_m/(s-r)^m$. Example: $1/s^2(s+1) = A/s+B/s^2+C/(s+1)$; $B=1$, $C=1$, $A=-1$; inverse: $f(t) = -1+t+e^{-t}$.

Case 3 (Irreducible quadratic): $P(s)/[(s^2+as+b)\cdot R(s)]$ includes term $(Cs+D)/(s^2+as+b)$, then complete the square. Example: $1/[(s^2+2s+5)(s+1)] = (As+B)/(s^2+2s+5)+C/(s+1)$; find constants, complete the square $(s+1)^2+4$, invert to $e^{-t}(\alpha\cos 2t+\beta\sin 2t)+\gamma e^{-t}$.

Q2. Solve completely: Find $\mathcal{L}^{-1}\left\{\dfrac{5s^2+6s+7}{(s+1)(s-2)(s+3)}\right\}$.

Answer: Decompose as $A/(s+1)+B/(s-2)+C/(s+3)$.

$$A = \frac{5(1)-6+7}{(-1-2)(-1+3)} = \frac{6}{-6} = -1.$$

$$B = \frac{5(4)+6(2)+7}{(2+1)(2+3)} = \frac{39}{15} = \frac{13}{5}.$$

$$C = \frac{5(9)+6(-3)+7}{(-3+1)(-3-2)} = \frac{34}{10} = \frac{17}{5}.$$

Verify at $s=0$:

$$\text{LHS} = \frac{7}{(1)(-2)(3)} = -\frac{7}{6}; \qquad \text{RHS} = \frac{-1}{1}+\frac{13/5}{-2}+\frac{17/5}{3} = -1-\frac{13}{10}+\frac{17}{15} = -\frac{30}{30}-\frac{39}{30}+\frac{34}{30} = -\frac{35}{30} =$$

$$f(t) = -e^{-t}+\frac{13}{5}e^{2t}+\frac{17}{5}e^{-3t}.$$

Q3. Solve completely: Find $\mathcal{L}^{-1}\left\{\dfrac{2s+5}{(s+1)^2(s+3)}\right\}$.

Answer: Decompose as $A/(s+1) + B/(s+1)^2 + C/(s+3)$.

$$B = \frac{2(-1)+5}{(-1+3)} = \frac{3}{2}.$$

$$C = \frac{2(-3)+5}{(-3+1)^2} = \frac{-1}{4}.$$

Expand for A: $2s+5 = A(s+1)(s+3) + B(s+3) + C(s+1)^2$. At $s = 0$: $5 = 3A+3B+C = 3A+9/2-1/4 \implies 3A = 5-9/2+1/4 = 20/4-18/4+1/4 = 3/4 \implies A = 1/4$. Check s^2 coefficient: $0 = A+C = 1/4-1/4 = 0.\checkmark$

$$f(t) = \frac{1}{4}e^{-t} + \frac{3}{2}te^{-t} - \frac{1}{4}e^{-3t}.$$

Q4. Solve completely: Find $\mathcal{L}^{-1}\left\{\dfrac{3s+2}{(s^2+2s+10)(s+4)}\right\}$.

Answer: Decompose as $(As+B)/(s^2+2s+10) + C/(s+4)$.

$$C = \frac{3(-4)+2}{16-8+10} = \frac{-10}{18} = -\frac{5}{9}.$$

Multiply through: $3s+2 = (As+B)(s+4) + C(s^2+2s+10)$. s^2: $0 = A+C \implies A = 5/9$. s^0: $2 = 4B+10C = 4B-50/9 \implies 4B = 2+50/9 = 68/9 \implies B = 17/9$. Complete square: $s^2+2s+10 = (s+1)^2+9$, $p = 1$, $q = 3$. Split: $(5s/9+17/9)/[(s+1)^2+9] = (5/9)(s+1)/[(s+1)^2+9]+(12/9)(1/3)\cdot 3/[(s+1)^2+9]$.

$$f(t) = \frac{5}{9}e^{-t}\cos(3t) + \frac{4}{9}e^{-t}\sin(3t) - \frac{5}{9}e^{-4t}.$$

Q5. Explain when polynomial long division is needed before partial fractions, and demonstrate with $F(s) = (s^3+3s^2+2s+4)/(s^2+1)$.

Answer: Long division is needed whenever $\deg P \geq \deg Q$, because partial fractions require a proper rational function. For $F(s) = (s^3+3s^2+2s+4)/(s^2+1)$:

$$s^3+3s^2+2s+4 = (s+3)(s^2+1) + (s+1)$$
$$\implies F(s) = s+3+\frac{s+1}{s^2+1}.$$

The polynomial part $s+3$ inverts to $\delta'(t)+3\delta(t)$. The proper part $(s+1)/(s^2+1) = s/(s^2+1)+1/(s^2+1)$ inverts to $\cos t+\sin t$. Therefore $f(t) = \delta'(t)+3\delta(t)+\cos t+\sin t$.

17 Practice Problems

P1. $\mathcal{L}^{-1}\left\{\dfrac{2s+1}{(s+1)(s+3)}\right\}$

Hint: Decompose as $A/(s+1)+B/(s+3)$. Use cover-up: $A = (-1)/2$, $B = 5/2$. Answer: $-\frac{1}{2}e^{-t} + \frac{5}{2}e^{-3t}$.

P2. $\mathcal{L}^{-1}\left\{\dfrac{1}{s(s^2+1)}\right\}$

Hint: Decompose as $A/s + (Bs + C)/(s^2 + 1)$. Set $s = 0$: $A = 1$. Compare s^1: $C = 0$, s^2: $A + B = 0 \implies B = -1$. Answer: $1 - \cos t$.

P3. $\mathcal{L}^{-1}\left\{\dfrac{s}{(s-1)(s^2+2s+5)}\right\}$

Hint: Decompose as $A/(s-1)+(Bs+C)/(s^2+2s+5)$. Cover-up at $s = 1$: $A = 1/(1+2+5) = 1/8$. Complete square: $(s+1)^2+4$. Answer: $\frac{1}{8}e^t+e^{-t}\big(-\frac{1}{8}\cos(2t) + \frac{3}{8}\sin(2t)\big)$.

P4. $\mathcal{L}^{-1}\left\{\dfrac{s^2+s+1}{s^2(s+1)}\right\}$

Hint: Decompose as $A/s+B/s^2+C/(s+1)$. $B = 1$ (set $s = 0$), $C = 1$ (set $s = -1$), $A = 0$ (compare s^2). Answer: $t + e^{-t}$.

P5. $\mathcal{L}^{-1}\left\{\dfrac{3}{(s^2+s)^2}\right\} = \mathcal{L}^{-1}\left\{\dfrac{3}{s^2(s+1)^2}\right\}$

Hint (approach): Decompose as $A/s + B/s^2 + C/(s + 1) + D/(s + 1)^2$. This is a repeated linear case at both $s = 0$ and $s = -1$. $B = 3$ (cover-up s^2, $s = 0$), $D = 3$ (cover-up $(s + 1)^2$, $s = -1$). Compare s^0: $3 = B + D +$ others — full system gives $A = -6$, $C = 6$. Answer: $-6 + 3t + 6e^{-t} + 3te^{-t}$.

P6. $\mathcal{L}^{-1}\left\{\dfrac{s^3+1}{s^2(s^2+4)}\right\}$

Hint: Check: $\deg P = 3 < \deg Q = 4$, so the fraction is proper. Decompose as $A/s + B/s^2 + (Cs + D)/(s^2 + 4)$. Comparing s^0: $1 = 4B \implies B = 1/4$. s^1: $0 = 4A \implies A = 0$. s^2: $0 = B + D \implies D = -1/4$. s^3: $1 = A + C \implies C = 1$. Answer: $\frac{1}{4}t + \cos(2t) - \frac{1}{8}\sin(2t)$.

18 MCQs

1. Before applying partial fractions to $F(s) = P(s)/Q(s)$, which condition must be verified?

(a) $\deg Q < \deg P$
(b) $\deg P = \deg Q$
(c) $\deg P < \deg Q$ **(fraction is proper)**
(d) $P(s)$ and $Q(s)$ share a common root

Explanation: Partial fractions are valid only for proper rational functions. If $\deg P \geq \deg Q$, polynomial long division must be performed first.

2. For $F(s) = P(s)/(s-a)^3$, how many unknown constants appear in the partial fraction decomposition?

(a) 1

(b) 2

(c) **3**

(d) Depends on $\deg P$

Explanation: A repeated factor $(s-a)^m$ always contributes exactly m terms in the decomposition, one for each power from 1 to m.

3. The correct partial fraction form for $P(s)/[(s^2+4)(s+1)]$ is:

(a) $A/(s^2+4)+B/(s+1)$

(b) $(As+B)/(s^2+4)+C/(s+1)$

(c) $(As^2+Bs+C)/(s^2+4)+D/(s+1)$

(d) $A/(s+2i)+B/(s-2i)+C/(s+1)$

Explanation: The irreducible quadratic s^2+4 requires a linear numerator $As+B$; using only a constant A is insufficient to represent the general partial fraction.

4. The Heaviside cover-up formula for finding constant A_k in the decomposition with distinct roots is:

(a) $A_k = P(s)/Q(s)$

(b) $A_k = Q'(r_k)/P(r_k)$

(c) $A_k = \lim_{s\to r_k} F(s)$

(d) $A_k = \lim_{s\to r_k}(s-r_k)F(s) = P(r_k)/\prod_{j\neq k}(r_k-r_j)$

Explanation: Multiplying by $(s-r_k)$ and setting $s=r_k$ eliminates all other terms, leaving A_k directly.

5. The denominator factor (s^2+4) requires which treatment in partial fractions?

(a) Factor it as $(s+2)(s-2)$ and use two linear terms

(b) **Treat it as an irreducible quadratic; use numerator $(As+B)$ and complete the square**

(c) Ignore it — only linear factors need decomposition

(d) Use a constant numerator A since the expression is already simple

Explanation: s^2+4 has discriminant $0-16<0$, so it has no real roots and is irreducible over the reals. Factoring as $(s+2)(s-2)$ is wrong because $s^2+4 \neq (s+2)(s-2) = s^2-4$.

19 Common Mistakes Box

Common Mistakes in Partial Fractions

Mistake	What Students Do	Correct Approach
Incomplete repeated root form	Write only $A/(s-r)^2$ for a double root	Use both $A_1/(s-r) + A_2/(s-r)^2$ — the full form needs one term for every power from 1 to m
Factoring s^2+4 incorrectly	Treat s^2+4 as $(s+2)(s-2)$ and use two linear terms	s^2+4 is irreducible over $\mathbb{R}$ ($s^2-4=(s+2)(s-2)$, not s^2+4); use $(As+B)/(s^2+4)$
Skipping polynomial long division	Apply partial fractions directly to an improper $F(s)$	Always check $\deg P$ vs. $\deg Q$ first; divide if $\deg P \geq \deg Q$ before decomposing
Constant numerator for quadratic	Write $A/(s^2+as+b)$ instead of $(As+B)/(s^2+as+b)$	A degree-2 denominator factor requires a degree-1 numerator to allow enough unknowns to match all powers
Not verifying constants	Accept A, B, C values without checking	Always substitute a convenient value of s (e.g. $s=0$) back into the identity to confirm the decomposition is correct

20 Quick Recap

Quick Recap — Method of Partial Fractions

- **Proper fraction check:** Always verify $\deg P < \deg Q$; if not, divide first.
- **Three main cases:** (1) Distinct real roots: $A_k/(s-r_k)$; (2) Repeated root $(s-r)^m$: m terms $A_1/(s-r) + \cdots + A_m/(s-r)^m$; (3) Irreducible quadratic: $(As+B)/(s^2+as+b)$.
- **Heaviside cover-up** finds each A_k for distinct linear roots in one step: $A_k = (s-r_k)F(s)\big|_{s=r_k}$.
- **Repeated root coefficients:** Highest power by cover-up on the whole block; lower powers by substitution or differentiating $(s-r)^m F(s)$ and evaluating at $s=r$.
- **Irreducible quadratic:** After finding A, B, complete the square $s^2+as+b = (s+p)^2+q^2$ and split numerator to match $e^{-pt}\cos(qt)$ and $e^{-pt}\sin(qt)$ table entries.
- **Systematic 6-step procedure:** (1) Check proper; (2) factor $Q(s)$; (3) write decomposition form; (4) find constants; (5) invert each term; (6) combine.
- **Total unknowns** always equals $\deg Q$ — use this as a sanity check before

solving.

- **Python/sympy verification:** `sympy.apart(F, s)` gives the decomposition symbolically; re-take the Laplace transform to confirm $f(t)$ is correct.